Module 2: Overfishing

```
library(tidyverse)
```

Use of GitHub

Link to your forked GH repository:

Use of Quarto

Link to your .qmd file:

Overexploitation of global fisheries

We will examine the state of global fisheries and seek to reproduce one of the most widely cited examples of species collapse in order to examine the evidence behind an influential paper on global fisheries, Worm et al. 2006. However, rather than using the 1950-2003 dataset available to Worm and colleagues in 2006, we will be drawing from more recent stock assessment data to see how the trends have fared, specifically in Atlantic cod.

We also explore how to understand and manipulate tabular data using relational database concepts. Instead of working with a single independent dataframe, as in Module 1, we will be working with a large relational database consisting of tables of different dimensions, but are related through one or more IDs.

The Database

We will use data from the RAM Legacy Stock Assessment Database. Note that the database files are available through Zenodo; please download and unzip the file, moving the .RData file only into your own `data/` folder within this project.

Use the code below to load up the tables in this database.

```
load("data/DBdata[asmt][v4.66].RData")
```

The above function loads all 60 tables of this database, which is a lot! So step one of the data science process is exploring the structure of the database and figuring out how the tables are related.

A selected set of documentation have been provided in `documentation/`. Open up the Database Quickstart Guide for reference as you work through the problems below.

Q1 (1 point)

One of the main metadata tables is 'stock':

- How many stocks are there and how many species do they represent? Either common or scientific names okay, although they give different answers.
- What are the top three most common stocks? (Okay to use common names; these are mostly standardized for fish)
- In your own words, describe the unit of observation in the 'stock' table and some associated variables that describe it.

```
stock_table = stock

#there are 1514 fish stocks
#view(stock_table)

stock_table |>
  select(commonname) |>
  unique() |>
  mutate(number = n())
```

```
                      commonname number
1                 Acadian redfish    410
5         Argentine chub mackerel    410
6                       Alfonsino    410
7                   Albacore tuna    410
13                  Alaska plaice    410
14                   Alaska skate    410
15                American plaice    410
22                        Anchovy    410
41                  Anchovy kilka    410
```

42	Black bellied angler	410
46	Anglerfish spp	410
49	Walleye pollock	410
53	Alaskan pink shrimp	410
54	Arctic surfclam	410
58	Arrowtooth flounder	410
61	Argentine anchoita	410
63	Argentine hake	410
65	Arabesque greenling	410
68	Amberstripe scad	410
69	Angel shark	410
70	Atlantic bluefin tuna	410
72	Atlantic halibut	410
75	Atka mackerel	410
77	Atlantic croaker	410
78	Antarctic toothfish	410
79	Aurora rockfish	410
80	Australian salmon	410
81	Atlantic wolffish	410
82	Axillary seabream	410
83	Blue crab	410
84	Bobo croaker	410
85	Beryx spp	410
86	Baltic flounder	410
87	Blue grenadier	410
89	Blackgill rockfish	410
90	Bigeye grunt	410
92	Benguela hake	410
93	Bonnethead shark	410
94	Bigeye tuna	410
98	Bight redfish	410
99	Big skate	410
105	Blue jack mackerel	410
106	Black cardinalfish	410
107	Blue king crab	410
109	Black scabbardfish	410
110	Black grouper	410
111	Black oreo	410
113	Black rockfish	410
118	Blue ling	410
121	Black marlin	410
122	Blonde ray	410
127	Blue shark	410

131	Blueline tilefish	410
132	Bluefish	410
133	Blue rockfish	410
134	Spotted mackerel	410
136	Blue marlin	410
137	Indo Pacific blue marlin	410
139	Blackmouth dogfish	410
141	Bank rockfish	410
142	Blacknose shark	410
143	Bluenose	410
144	Boarfish	410
145	Bocaccio	410
147	Bonga	410
149	Brill	410
151	Yellow sea bream	410
152	Seabream	410
155	Brown rockfish	410
156	Brown shrimp	410
157	Black sea bass	410
159	Bluespotted seabream	410
160	Black scraper	410
161	Barndoor skate	410
162	Basking shark	410
163	Yellow squat lobster	410
166	Blacktip shark	410
168	Brown tiger shrimp	410
169	Atlantic butterfish	410
170	Blue whiting	410
174	Beni zuwai crab	410
175	Cabezon	410
179	California scorpionfish	410
180	Capelin	410
184	Catfish	410
187	Shallow water cape hake	410
188	Chilean herring	410
189	Chilipepper	410
190	Cunene horse mackerel	410
193	China rockfish	410
196	Cape Hope squid	410
197	Chilean jack mackerel	410
198	Crimson jobfish	410
200	Cuckoo ray	410
204	Atlantic chub mackerel	410

205	Pacific chub mackerel	410
206	Chub mackerel	410
208	Cobia	410
210	Atlantic cod	410
238	Cowcod	410
239	Common pandora	410
241	Copper rockfish	410
242	West coast rock lobster	410
247	Croaker	410
251	Canary rockfish	410
253	Chum salmon	410
352	Chilean sardine	410
353	Coastal shrimp	410
354	Clearnose skate	410
355	Caribbean spiny lobster	410
358	Cape horse mackerel	410
359	Cusk	410
360	Tusk	410
365	Cuttlefish	410
369	Dab	410
371	Deep water cape hake	410
372	Deepwater flathead	410
373	Darkblotched rockfish	410
374	Dredge oyster	410
375	Deep water rose shrimp	410
376	Dusky shark	410
377	Dover sole	410
379	Deep sea smelt	410
381	Dusky rockfish	410
382	Devil anglerfish	410
384	European seabass	410
389	East coast sole	410
390	European flounder	410
395	European pollack	410
397	English sole	410
399	European squid	410
402	Eulachon	410
405	Flathead flounder	410
406	Flathead sole	410
408	Fourspotted megrim	410
410	False scad	410
411	Flame snapper	410
412	Finetooth shark	410

413	Gag	410
415	Common gemfish	410
417	Greater forkbeard	410
419	Grey grunt	410
421	Grey gurnard	410
422	Greenland halibut	410
428	Gilthead seabream	410
429	Green jobfish	410
430	Golden king crab	410
434	Goliath grouper	410
435	Golden redfish	410
437	Gopher rockfish	410
438	Giant Pacific octopus	410
440	Greater amberjack	410
442	Greenstriped rockfish	410
443	Giant red shrimp	410
449	Gray snapper	410
450	Greenspotted rockfish	410
452	Greater silver smelt	410
456	Giant stargazer	410
457	Grooved tiger prawn	410
458	Giant trevally	410
459	Gray triggerfish	410
460	Green sea urchin	410
461	Haddock	410
475	Hake	410
487	Hake spp	410
489	Herring	410
503	Pacific herring	410
522	Horsehead tilefish	410
523	Horse mackerel	410
528	Hogfish	410
530	Hoki	410
532	Northern shortfin squid	410
533	Iceland scallop	410
534	Indo-Pacific sailfish	410
535	Japanese amberjack	410
536	Japanese anchovy	410
539	Japanese flying squid	410
541	Japanese jack mackerel	410
543	Japanese pufferfish	410
545	Japanese snapper	410
546	Kichiji rockfish	410

549	Kelp greenling	410
550	Kitefin shark	410
551	Kingklip	410
554	King mackerel	410
556	Kamchatka flounder	410
557	Longfin codling	410
558	Large eye dentex	410
560	Leafscale gulper shark	410
561	Largehead hairtail	410
562	Lingcod	410
565	Ling	410
569	Longfin inshore squid	410
570	Lavender jobfish	410
571	Longnose skate	410
577	Lane snapper	410
578	American lobster	410
591	Lesser spotted dogfish	410
595	Little skate	410
597	Lemon sole	410
598	Longspine thornyhead	410
599	Lumpfish	410
600	Atlantic mackerel	410
601	Mackerel	410
603	Megrim	410
604	Megrim spp	410
607	Gulf menhaden	410
608	Atlantic menhaden	410
609	Mediterranean horse mackerel	410
610	Monkfish	410
614	Jackass morwong	410
617	Madeiran sardinella	410
619	Sea mullet	410
620	Mutton snapper	410
621	Norway lobster	410
659	Norway pout	410
661	Northern rockfish	410
663	Northern rock sole	410
665	Nylon shrimp	410
675	New Zealand snapper	410
679	Octopus	410
683	Japanese flounder	410
687	Orange-footed sea cucumber	410
691	Ocean pout	410

692	Orange roughy	410
701	Oceanic whitetip shark	410
702	Pacific bluefin tuna	410
703	Peruvian anchoveta	410
710	Northern shrimp	410
713	Ocean shrimp	410
734	Pandora spp	410
735	Patagonian cod	410
736	Patagonian grenadier	410
738	Paua	410
743	Pink cusk eel	410
747	Pacific cod	410
758	Pacific ocean perch	410
761	Pacific geoduck	410
762	Pacific hake	410
763	Pacific halibut	410
764	Pointhead flounder	410
766	Picarel	410
767	Japanese sardine	410
769	Pink shrimp	410
770	European plaice	410
781	Pollock	410
785	Saithe	410
791	Porbeagle shark	410
793	Poor cod	410
794	Pacific red gurnard	410
796	Pink salmon	410
966	Pacific sandlance	410
968	Pacific sand lance	410
970	Petrale sole	410
971	Patagonian toothfish	410
974	Quillback rockfish	410
976	Ocean quahog	410
977	Red seabream	410
984	Red deepsea crab	410
985	Red shrimp	410
991	Deepwater redfish	410
992	Beaked redfish	410
998	Redfish	410
999	Redfish species	410
1006	Rex sole	410
1008	Rougheye rockfish	410
1010	Red grouper	410

1012	Rubberlip grunt	410
1013	Red gurnard	410
1014	Red hake	410
1016	Round herring	410
1018	Roughhead grenadier	410
1019	Red king crab	410
1023	Red mullet	410
1036	Roundnose grenadier	410
1040	Rock crab	410
1042	Red pandora	410
1044	Red porgy	410
1045	Red rock lobster	410
1052	Rock lobster	410
1054	Royal red shrimp	410
1055	Round sardinella	410
1057	Round nose flounder	410
1068	Rosette skate	410
1069	Red snapper	410
1071	Rock sole	410
1074	Red squat lobster	410
1077	Redstripe rockfish	410
1079	Roughscale sole	410
1080	South African abalone	410
1081	Sablefish	410
1084	Sailfish	410
1086	Pacific sardine	410
1087	Sardinella	410
1090	European pilchard	410
1091	Sardine	410
1111	Pacific saury	410
1112	Sandbar shark	410
1113	Shortbelly rockfish	410
1114	Seabreams spp	410
1115	Deepwater cardinalfish	410
1116	Southern bluefin tuna	410
1117	Southern blue whiting	410
1120	Scad spp	410
1121	Sea scallop	410
1129	Scalloped hammerhead	410
1130	Scampi	410
1134	Scup	410
1135	Spiny dogfish	410
1138	Sandy ray	410

1139	Sand eel	410
1147	Smalleyed ray	410
1149	Summer flounder	410
1150	Shortfin mako	410
1154	Shagreen ray	410
1155	Silver hake	410
1158	Shako	410
1159	Sharpchin rockfish	410
1160	Smooth hound	410
1161	Silverfish	410
1162	Skipjack tuna	410
1167	Silk snapper	410
1168	Silky shark	410
1169	Smooth dogfish	410
1170	Smooth oreo	410
1175	Smooth skate	410
1177	Spottail mantis shrimp	410
1181	Surmullet	410
1184	Snapper	410
1190	Atlantic sharpnose shark	410
1191	Snow crab	410
1210	Snowy grouper	410
1211	Splitnose rockfish	410
1212	Common sole	410
1213	Sole	410
1228	Southern hake	410
1232	Japanese Spanish mackerel	410
1233	Spanish mackerel	410
1236	South Pacific hake	410
1237	Sprat	410
1243	Spotted ray	410
1249	Spotted spiny dogfish	410
1250	Southern pink shrimp	410
1253	Spear squid	410
1255	Spurdog	410
1256	Shortraker rockfish	410
1258	Sockeye salmon	410
1362	Sailfin sandfish	410
1364	Southern sardine	410
1365	Softshell clam	410
1366	Sidestripe shrimp	410
1369	Southern spiny lobster	410
1375	Shortspine thornyhead	410

1377	Starry flounder	410
1379	Striped marlin	410
1384	Starry ray	410
1385	Striped bass	410
1386	Striped red mullet	410
1388	Sea trout	410
1389	Aesop shrimp	410
1393	Swordtip squid	410
1394	Sleek unicornfish	410
1395	Atlantic surfclam	410
1397	School whiting	410
1398	Swordfish	410
1405	Tanner crab	410
1406	Tarakihi	410
1408	Tasmanian giant crab	410
1409	Thornback skate	410
1416	Threadfin	410
1419	Thresher shark	410
1420	Tiger flathead	410
1421	Tilefish	410
1424	Tope shark	410
1425	Trevally	410
1426	Twosaddle goatfish	410
1427	Thorny skate	410
1430	Turbot	410
1434	Undulate ray	410
1436	Vermilion snapper	410
1438	White anglerfish	410
1441	Blue warehou	410
1443	Weakfish	410
1444	White grouper	410
1445	White hake	410
1451	Whiting	410
1459	Windowpane flounder	410
1461	Winter flounder	410
1465	Witch flounder	410
1470	Willowy flounder	410
1471	White marlin	410
1482	Widow rockfish	410
1484	Whitespotted conger	410
1485	Whitesaddle goatfish	410
1486	White shrimp	410
1487	Winter skate	410

```
1489          Weathervane scallop    410
1490                  Waved whelk    410
1491           Yellowedge grouper    410
1492          Yellowtail flounder    410
1497           Yelloweye rockfish    410
1500           Yellowfin goatfish    410
1501               Yellowfin tuna    410
1505             Yellow goosefish    410
1506             Yellownose skate    410
1509      Yellow striped flounder    410
1511        Yellowstripe goatfish    410
1512               Yellowfin sole    410
1513          Yellowtail rockfish    410
1514           Yellowtail snapper    410
```

```
# seems like 410 common names are involved

stock_table |>
  group_by(commonname) |>
  select(commonname) |>
  mutate(num_stock = n()) |>
  unique() |>
  arrange(desc(num_stock))
```

```
# A tibble: 410 x 2
# Groups:   commonname [410]
   commonname      num_stock
   <chr>               <int>
 1 Pink salmon           170
 2 Sockeye salmon        104
 3 Chum salmon            99
 4 Norway lobster         38
 5 Atlantic cod           28
 6 Herring                24
 7 Anchovy                19
 8 Snow crab              19
 9 Northern shrimp        17
10 Walleye pollock        14
# i 400 more rows
```

```
# seems like there are the most Pink, Sockeye, and Chum salmon stocks
```

1514 fish stocks are included in the data, representing 410 species assuming each common name corresponds to one scientific name. Pink, Sockeye, and Chum salmon have the most stocks with 170, 104, and 99 respectively.

Each row in the stock table includes a identifier for the stock that includes some information about the country the stock resides in, the taxonomic serial number for which kind of fish is present, both the scientific and common names, an area id including the country the stock is in as well as the stock's position inside that country, a more detailed identifier of which country and region the stock belong to, a unique stock id, another area id, and a indicator of whether or not a stock is current / deprecated.

Q2 (1 point)

The next main metadata table is 'assessment':

- What is the unit of observation in the 'assessment' table? How is this related to the unit of observation in the 'stock' table?
- It seems like 'stockid' and 'assessyear' might uniquely identify each assessment. Is this the case? If not, why not?
- Which variable(s) might you use to join 'stock' and 'assessment'?

```
# entry 689 was invalid and making df unable to be read

assessment[] = lapply(assessment, function(x) {
  if (is.character(x)) iconv(x,
                             from = "latin1",
                             to = "UTF-8") else x })


view(assessment)
```

Each entry in "assessment" records several identifiers for a given stock, the location of the stock, date of recording / loading along with the year they happened, the method used to assess a given stock (if any), and a link to the excel sheet where the data are located. There are several other columns included, however the /documentation sheet does not indicate what they are and most of their values are either "null" or the same across most observations, so I have ignored them.

Let's check to see if stockid and assessyear are truly unique across all rows:

```
# groups by stock id then counts. count should be 1 if stockid is unique

unique_stockid = assessment |>
  group_by(stockid) |>
  mutate(num_unique_stockid = n()) |>
  filter(num_unique_stockid > 1) |>
  ungroup()

view(unique_stockid)
```

Alright, so stockid is not truly unique – there are 2652 observations out of 3497 that share the same id with at least one other observation.

```
unique_assessyear = assessment |>
  group_by(assessyear) |>
  mutate(num_unique_assessyear = n()) |>
  filter(num_unique_assessyear > 1) |>
  ungroup()

view(unique_assessyear)
```

Okay so assessyear (as expected) is not unique with 3061 of 3497 entries sharing the same year with at least one other observation.

Let's see if stockid + assessyear together works:

```
unique_stkid_asyear = assessment |>
  group_by(stockid, assessyear) |>
  mutate(num_unique = n()) |>
  filter(num_unique > 1)

view(unique_stkid_asyear)
```

Again, both columns together are not unique since 96 out of the 3497 total entries share both stockid and assessyear with at least one other observation.

let's check which columns are shared between stock and assessment:

```
intersect(colnames(stock_table), colnames(assessment))
```

```
[1] "stockid"   "stocklong"
```

```
# stockid and stocklong are the only ones shared
# lets see if they're unique for both dataframes:

stock_table |>
  group_by(stockid, stocklong) |>
  mutate(num_unique = n()) |>
  filter(num_unique > 1) |>
  length()
```

```
[1] 16
```

```
assessment |>
  group_by(stockid, stocklong) |>
  mutate(num_unique = n()) |>
  filter(num_unique > 1) |>
  length()
```

```
[1] 19
```

```
# so grouping by both stockid and stocklong is not enough to get a unique key for each datafi
```

The columns that are shared between stock and assessment (stockid / stocklong) are not unique when paired together in either table. The stock table has another identifier "GRSF_uuid" that seems to be unique or almost unique, so if it were possible to add this column to assessment, I think both dataframes could be joined without a problem. However, as they are currently, I don't think it is reasonable to join them.

Q3 (1 point)

In section B of the Database Quick Guide, additional metadata tables are listed in section B. With your data exploration skills, complete the following table:

B table	Associated table	join column from B	join column from other
area	management	areatype	mgmt
assessmethod	assessment	assessmethod	method_short
assessor	management	mgmt	mgmt
management	area, assessor	mgmt	areatype, mgmt respectively

B table	Associated table	join column from B	join column from other
taxonomy	stock	tsn	tsn

*Note that assessmethod is not a data.frame, which was perhaps an oversight; consider turning it into one for the ease of using tidyverse functions.

```
assessmethod_df = as_tibble(assessmethod)


assessmethod_df = assessmethod_df |>
  rename(category = 1,
         abrev_method = 2,
         full_method_name = 3)



#for loop for checking intersection that didn't work

#list_of_b_tables = list(area,assessmethod_df,assessor,management,taxonomy)

#vector = vector(mode = "list", length = 100)

#x = 20

#for (first_table in list_of_b_tables) {
 # for (second_table in list_of_b_tables) {
  #  c(paste("col names shared between",
   #   deparse(substitute(first_table)),
    #  "and",
     # # deparse(substitute(second_table)),": ",
      #intersect(colnames(first_table),
       #       colnames(second_table))))
    #


   # }
#}


view(area)
```

```
view(assessmethod_df)
view(assessor)
view(management)
view(taxonomy)
```

Q4 (1 point)

There are two main data tables described in section C, but their key attributes are difficult to understand. Join the two main data tables with their respective metrics tables (section D):

- Briefly describe the contents of each joined table.
- What is accomplished by joining these tables?

Overexploitation of Atlantic cod

This database is clearly complex and contains a lot of information. We've grappled with its high dimensionality by exploring and joining tables above. Next, we will take a "bottom-up" approach and evaluate a single species, Atlantic cod. Read this book chapter on Atlantic cod, which serves as a reference for the following questions.

Q5 (1 point)

The collapse of the Atlantic cod fisheries is well documented. Let's see if we can visualize that decline with this database.

- How many unique fish stocks comprise "Atlantic cod" and what regions do they come from?

Create a vector of the 'stockid' associated with "Atlantic cod". Then turn your attention to the previously-joined timeseries/tsmetrics table and create a subsetted dataframe with only observations of "Atlantic cod" that have a non-NA value. This table reports many years and many metrics, while we are primarily interested in the years since 1950.

- Which 2 metrics are most commonly reported for Atlantic cod in this timeframe and what do they mean?

Q6 (1 point)

Make a timeseries figure for each stock of Atlantic cod in a single figure. Rather than coloring by the identity of the stock, color by the region it comes from (this may require joining). Use `facet_*()` to plot the top two metrics from Q5 in a single code chunk. Connect the points for each stock and focus on the period since 1950. Update until the figure is easily interpretable.

- What happened to Atlantic cod stocks during the period 1950-present? What information does each metric convey?

Q7 (1 point)

We will recreate a slightly more complex version of a figure from the Millennium Ecosystem Assessment (Fig. 11).

Selecting a metric that can be added across stocks, calculate the total fish landings for each region of Atlantic cod. Make a plot similar to Fig. 11 and color by region.

Which region does Fig. 11 from the MEA correspond to?

Q8 (1 point)

Under the Management Policies section of the book chapter, the authors summarize a few major efforts to restore and protect stocks of Atlantic cod. Focusing only on the US and Canada, update your plot from above to include vertical lines to represent the timing of least three major events and label them clearly on your plot. Doing so will likely involve creating a separate dataframe and using it in one of your ggplot layers (one possible argument within `aes()` is `linetype`). Facet the plot by region and update until it is easily interpretable.

Q9 (2 points)

What happened to Atlantic cod stocks for each region following each piece of legislation or management plan, and why do you think this was so? Your multi-paragraph answer should be informed by content from the book chapter and your figure from Q8; additional sources are optional, and all sources should be cited.